Fiji Revenue and Customs Service (FRCS)

Configuration Manual

Version 1.0.0 – April 2026

A Free and Open Source Point of Sale System for MSMEs in Fiji

Developed by

FRCS – IT Team

Licensed as free and open-source software

1.0 Introduction

The Fiji Revenue and Customs Service (FRCS), in collaboration with the Fiji Revenue and Customs Service (FRCS), developed the FreePOS system – a secure, VAT-compliant Point of Sale system for Micro, Small, and Medium Enterprises (MSMEs) across Fiji. The system is built on Odoo 18 Community Edition and integrates with the FRCS TaxCore V-SDC API for real-time fiscal receipt generation.

This Configuration Manual covers the business-level setup of the FreePOS system after successful installation on a Windows machine running Odoo 18 directly (without Docker or Caddy). It includes module activation, Fiji-specific tax settings, POS shop setup, payment methods, TaxCore integration, custom login page branding, and hardware configuration.

NOTE: This manual assumes Odoo 18 Community Edition has been installed on Windows and the custom modules have been placed in the addons directory. No Docker, Caddy, or reverse proxy setup is required for this version.

This manual is intended for business owners and system administrators responsible for configuring and maintaining the FreePOS system.

2.0 System Architecture Overview

The FreePOS system runs directly on a Windows PC using Odoo 18 Community Edition. There is no Docker container or web proxy required. The system communicates with the FRCS TaxCore V-SDC API over HTTPS for fiscal invoice signing.

Component	Description
Operating System	Windows 10 / Windows 11 (64-bit)
Application Server	Odoo 18 Community Edition (Windows installer)
Database	PostgreSQL 15+ (bundled with Odoo installer)
Python	Python 3.12 (bundled with Odoo installer)
Web Access	http://localhost:8069 or http://pos.local (with hosts file)
TaxCore	FRCS V-SDC API – https://vsdc.sandbox.vms.frcs.org.fj
Receipt Printer	USB ESC/POS receipt printer (connected directly to PC)
Barcode Scanner	USB HID barcode scanner (plug-and-play)

3.0 Initial Odoo Configuration

The FreePOS includes four custom Odoo modules developed specifically for Fiji FRCS requirements. Before configuring the system for daily use, these modules must be installed in the correct order.

3.1 Pre-Configuration Checklist

Before performing any configuration ensure the following have been completed:

- Odoo 18 Community Edition is installed and running on Windows
- PostgreSQL database has been created and initialized
- All four custom module folders are placed in the Odoo addons directory:
  - C:\Program Files\Odoo 18.0.XXXXXXXX\server\odoo\addons\
- Odoo service is running and accessible at http://localhost:8069
- You are logged in as Administrator

3.2 Custom Module Overview

3.2.1 FRCS POS (custom_pos)

Extends the standard Odoo POS with Fiji-specific features including TaxCore V-SDC signing, additional sale types (Proforma, Training, Copy, Advance Sale), fiscal receipt templates, and refund handling. This module must be installed FIRST.

3.2.2 Fiji Mini Chart of Accounts (l10n_fj_minicoa)

Provides the Fiji-specific localization including the Mini Chart of Accounts (MiniCOA) for MSMEs, Fiji VAT rates (0%, 9%, 12.5%, 15%), FJD currency setup, tax groups, and fiscal positions. The POS MiniCOA bootstrap module (pos_minicoa) installs automatically alongside this module.

3.2.3 FRCS Inventory (frcs_inventory)

Extends Odoo inventory with GTIN (Global Trade Item Number) support, product expiry tracking, improved product configuration, and an inventory dashboard designed for Fiji MSMEs.

3.2.4 FRCS Purchase Order (frcs_pos_purchase)

Extends Odoo purchasing with vendor FRCS TIN fields, VAT label mapping for purchase order lines, and FRCS-compliant purchase reports. Supports downstream fiscalization by ensuring vendor data is consistent.

3.2.5 Custom Login (custom_login)

Replaces the default Odoo login page with a branded login screen. Allows administrators to configure a custom background image, business logo, layout style, and optional welcome motto. All settings are managed from the Odoo Settings menu and take effect immediately.

3.3 Custom Module Installation

Navigate to the Odoo Apps screen and remove the Apps filter to see all modules. Then search for FRCS to find the custom modules.

NOTE: Modules MUST be installed in the exact order listed below. Installing out of order may cause dependency errors or missing payment methods.

Installation Order:

1. Install FRCS POS Module (custom_pos)
2. Install Fiji Mini Chart of Accounts (l10n_fj_minicoa)
3. Install FRCS Inventory (frcs_inventory)
4. Install FRCS Purchase Order (frcs_pos_purchase)
5. Install Custom Login Module (custom_module)

The FRCS POS Mini COA bootstrap module (pos_minicoa) will automatically install alongside the Fiji Mini Chart of Accounts module. Click Activate on each module in the order above and wait for each to fully complete before proceeding to the next.

3.4 Custom Module Configuration

3.4.1 Fiji Localization (l10n_fj_minicoa)

After installing all modules, several configuration steps are required to ensure Fiji VAT rules, the Mini Chart of Accounts, and FJD currency are correctly applied.

Configure Company Information

6. Navigate to Settings → Users & Companies → Companies
7. Select your company (e.g., My Company)
8. Update the following fields:
   - Country: Fiji
   - Currency: FJD – Fijian Dollar
   - Company Name: Enter your registered business name
   - Tax ID: Enter your FRCS TIN number
9. Click Save

NOTE: If FJD does not appear in the currency dropdown, go to Accounting → Configuration → Currencies, find FJD and toggle it to Active, then return to company settings.

Apply Fiji Localization

10. Go to Settings → Invoicing (or Accounting)
11. Find the Fiscal Localization section
12. Click the Package dropdown and select Fiji – FRCS MiniCoA
13. Click Save

This activates:

- The Fiji Mini Chart of Accounts
- Fiji VAT templates (0%, 9%, 12.5%, 15%)
- Tax groups and fiscal positions
- Fiji Dollar (FJD) currency rules
- Default fiscal settings appropriate for local MSMEs

Verify Fiji VAT Rates

14. Go to Invoicing → Configuration → Taxes

Ensure the following VAT rates are active (each has both Sales and Purchase versions):

- VAT 0% (Sales and Purchase)
- VAT 12.5% (Sales and Purchase)

NOTE: VAT 9% and 15% are also present in the system but may be inactive. Only 0% and 12.5% are required for standard FRCS compliance.

Verify Tax Groups

15. Navigate to Invoicing → Configuration → Tax Groups

Ensure the FRCS VAT tax group is present with the correct accounts:

- Tax Payable Account: 213100 VAT Collected
- Tax Receivable Account: 213300 VAT Paid

Confirm Journals

16. Go to Invoicing → Configuration → Journals

Ensure the following journals are present:

- POS Cash (Cash type)
- POS Card (Bank type)
- POS Mobile Money (Bank type)
- POS Wire Transfer (Bank type)
- POS Check (Bank type)

Verify Payment Methods

17. Go to Point of Sale → Configuration → Payment Methods

The following payment methods should be automatically created:

Payment Method	Purpose
POS Cash	Standard cash payments at the register
POS Cash 2	Second cash register (if two POS registers are used)
POS Card	EFTPOS / Debit / Credit card payments
POS Mobile Money	M-PAiSA, MyCash and other mobile wallet payments
POS Wire Transfer	Bank wire transfer payments
POS Check	Cheque payments

NOTE: The POS system is not integrated directly with EFTPOS terminals or mobile wallet apps. Payment methods are for record-keeping and TaxCore fiscalization purposes only. Each cash method is linked to a specific POS register – two cash registers require two separate cash payment methods.

3.4.2 Point of Sale Configuration

After all modules are installed, configure your POS shop for daily operation.

Creating a New Point of Sale Shop

18. Log in as Administrator
19. Navigate to Point of Sale → Configuration → Point of Sales
20. Click New to create a new POS shop
21. Enter a suitable name (e.g., Main Store)
22. Click Save

Adding Payment Methods to POS

23. Open your POS configuration
24. Go to the Payment section
25. Add the payment methods applicable to your register:
    - POS Cash (required – one per register)
    - POS Card
    - POS Mobile Money
    - POS Wire Transfer
    - POS Check
26. Click Save

NOTE: If you have two registers, assign POS Cash to Register 1 and POS Cash 2 to Register 2. Card, Mobile Money, Wire Transfer and Check can be shared across all registers.

3.4.3 Custom Login (custom_login)

The Custom Login module allows you to brand the FreePOS login screen with your business logo, a custom background image, and an optional welcome motto. All settings are managed from the Odoo Settings menu and take effect immediately without a system restart.

Accessing the Custom Login Settings

27. Navigate to Settings → General Settings
28. Scroll to the Custom Login section
29. Configure the options below as needed and click Save

Available Settings

Setting	Description
Login Page Design	Layout style for the login form. Options: Fullscreen Right, Fullscreen Left, Boxed Left, Boxed Right, Boxed Center.
Login Page Custom Background	Enable to use a custom uploaded image as the page background instead of the default solid colour.
Login Page Background Image	Upload an image file (PNG or JPG) to use as the login page background when custom background is enabled.
Login Page Show Logo	Enable to display your company logo on the login page.
Login Page Logo	Upload your company logo image (PNG or JPG) to display on the login page.
Show Motto	Enable to display a motivational or welcome quote on the left side of the login page.
Motto Text	The text to display as the welcome motto or quote on the login page.
Motto Text Color	Colour of the motto text (hex code, e.g. #FFFFFF for white). Defaults to black if left empty.

3.4.4 Email Configuration

FreePOS supports sending fiscal receipts directly to a customer's email address from the receipt screen after a completed transaction. Before this feature can be used, an outgoing mail server must be configured in Odoo.

Step 1 – Configure Outgoing Mail Server

Navigate to **Settings** → **Technical** → **Email** → **Outgoing Mail Servers** and click **New**. Enter the following details and click **Test Connection** before saving:

Field	Value
Name	Office 365 SMTP
SMTP Server	smtp.office365.com
SMTP Port	587
Connection Encryption	TLS (STARTTLS)
Username	your-email@yourdomain.com
Password	Your Microsoft 365 password

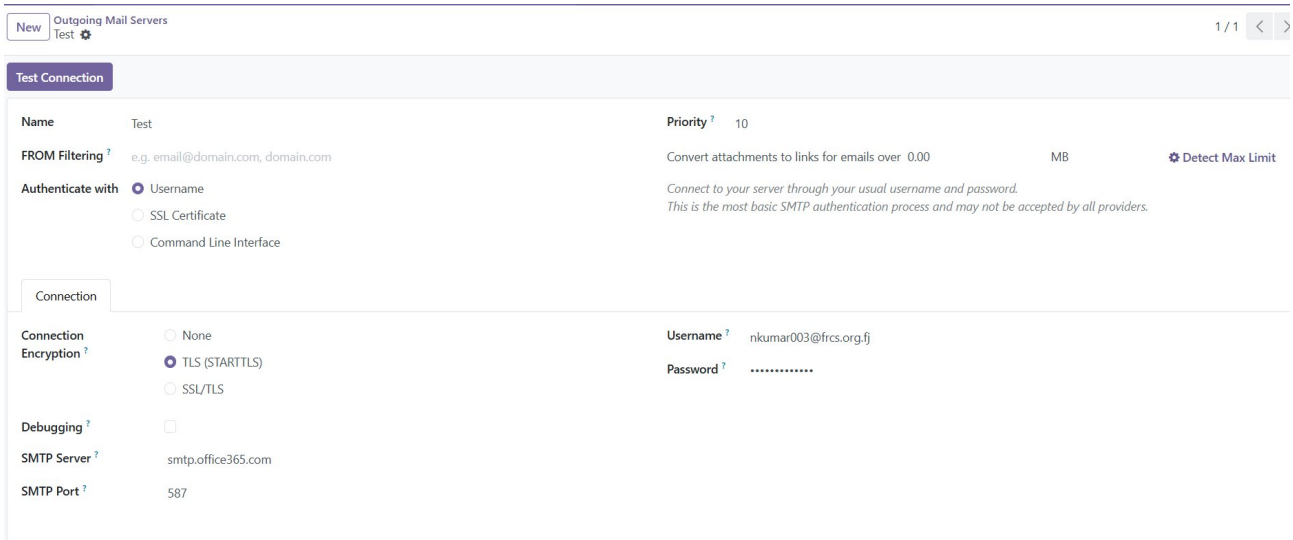

Figure 1 – Outgoing Mail Server configuration screen in Odoo Settings

NOTE: If using a Microsoft 365 account, SMTP AUTH must be enabled for your mailbox by your IT administrator before this will work. If the test connection fails with an authentication error, contact your IT team and ask them to enable SMTP AUTH for your account in the Microsoft 365 Admin Center. Refer to: https://aka.ms/smtp_auth_disabled

Step 2 – Sending a Receipt by Email

After a transaction is completed, the receipt screen is displayed. To email the receipt to the customer:

1. On the receipt screen, locate the email input field below the **Print Full Receipt** button.
2. Type the customer's email address into the field (e.g. john.doe@mail.com).
3. Click the **Send** button (paper plane icon) to the right of the field.

4. The fiscal receipt will be emailed to the customer immediately.

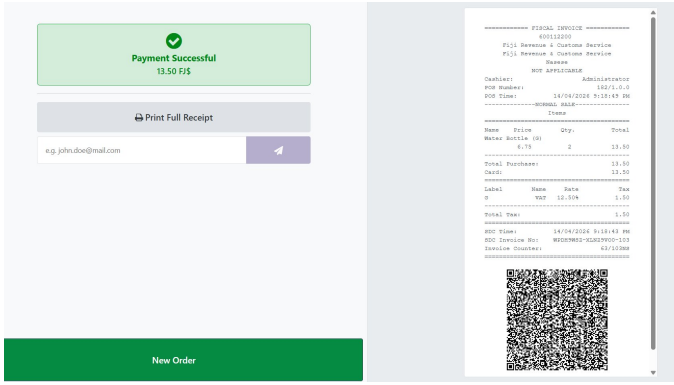

Figure 2 – Receipt screen showing the email input field and fiscal receipt preview

NOTE: The email input field is visible on the receipt screen by default. No additional POS configuration is required once the outgoing mail server has been set up. The emailed receipt includes the full fiscal invoice details including the SDC Invoice Number, Invoice Counter, and QR verification code.

The FreePOS integrates with the FRCS TaxCore V-SDC system for fiscal receipt signing. This configuration MUST be completed before using the POS. Valid credentials must be obtained from FRCS before proceeding.

NOTE: TaxCore credentials (PFX certificate, password, and PAC code) are provided by FRCS upon business registration. Contact FRCS to obtain these before attempting this configuration.

Required Information from FRCS:

- PFX Certificate file (.pfx format)
- Certificate password
- PAC code (Private Access Code)
- V-SDC API URL (sandbox: https://vsdc.sandbox.vms.frcs.org.fj)

Configuration Steps:

30. Navigate to Point of Sale → FRCS → V-SDC Settings
31. Click New to create a new configuration
32. Set the Company field to your company
33. Enter the V-SDC API URL provided by FRCS
34. Enter your PAC code
35. Under the Certificate section, upload your .pfx certificate file
36. Enter the PFX password
37. Click Save
38. Click the Convert PFX button to extract the certificate – the Cert Status field should show Active

NOTE: Only the Administrator can configure V-SDC settings. Without a valid certificate the POS will be unable to generate fiscal invoices and all sales will fail TaxCore validation.

Verify TaxCore Connection:

After saving, process a test transaction from the POS. Open the browser developer console (F12 → Console) and check for:

- TaxCore response → should show invoiceNumber, sdcDateTime, and invoiceCounter
- No error messages related to certificate or connection

Field	Example Value
V-SDC API URL	https://vsdc.sandbox.vms.frcs.org.fj
PAC	3AYVNZ

PFX Certificate	BA2PAXC4-DeveloperAuthenticationCertificate.pfx
PFX Password	(provided by FRCS)
Cert Status	Active (after Convert PFX)

4.1 V-SDC Connection Status Badge

Every time a cashier opens a POS session, FreePOS automatically performs a mutual TLS authentication handshake with the TaxCore V-SDC in the background. The outcome is displayed as a live status badge in the top-right corner of the POS navigation bar.

This satisfies the FRCS certification requirement: *"Upon operation start, the POS and V-SDC are mutually authenticated."*

How It Works

1. When the POS session opens, FreePOS calls the V-SDC **/api/v3/tax-rates** endpoint using the configured client certificate (mTLS).
2. If the V-SDC responds correctly, mutual authentication is confirmed — both sides have verified each other's identity.
3. The badge in the navbar updates immediately to reflect the result.
4. The cashier is **not blocked** from using the POS while this check runs — authentication happens in the background.

Badge States

The badge appears in the POS top navigation bar and has three possible states:

Badge Appearance	Status	Meaning
Grey ● Spinning circle	Checking	Authentication is in progress. Appears briefly on session open.
Green ✓ V-SDC	V-SDC Connected	Authentication successful. The POS is ready to issue fiscal invoices.
Red ! V-SDC	V-SDC Unreachable	Authentication failed. Fiscal invoice signing will not work until resolved.

Tip: Hover your mouse over the badge to see the full status message (e.g. "V-SDC Connected" or "V-SDC Unreachable").

What to Do If the Badge Is Red

A red badge means FreePOS could not reach or authenticate with the V-SDC. Fiscal invoice signing will fail for all transactions until the issue is resolved. Check the following:

- **Certificate not converted:** Go to **Point of Sale** → **FRCS** → **V-SDC Settings**, open your configuration, and confirm the **Cert Status** field shows *Certificate extracted and ready*. If not, click **Convert PFX**.
- **Wrong PAC or V-SDC URL:** Verify the URL and PAC code match the values provided by FRCS exactly.
- **No internet / network issue:** Confirm the PC can reach *https://vsdc.sandbox.vms.frcs.org.fj* from a web browser.
- **Certificate expired:** Contact FRCS to obtain a renewed PFX certificate and repeat the configuration steps in Section 4.0.

NOTE: The badge authentication runs in the background and does not block the cashier from opening the POS. However, if the badge remains red, **do not process sales** — all fiscal invoice signing will fail and transactions will not be recorded on the FRCS V-SDC portal. Resolve the connection issue before beginning the trading day.

5.0 Hardware Configuration

The FreePOS supports USB receipt printers and USB barcode scanners. Both are optional but recommended for efficient operation.

5.1 Barcode Scanner Integration

USB barcode scanners are plug-and-play and require no Odoo configuration.

39. Plug the scanner into a USB port on the Windows PC
40. Windows will automatically install drivers
41. The scanner acts as a keyboard input device
42. Open the POS session and scan any product barcode

The POS will automatically:

- Find the matching product by barcode or GTIN
- Add it to the current order
- Increase quantity if the product is already in the cart

NOTE: Products must have their GTIN or Barcode field filled in under Inventory → Products → Product Information tab for barcode scanning to work correctly.

5.2 USB Printer Configuration

The FreePOS supports ESC/POS USB receipt printers for automatic fiscal receipt printing after each transaction. The printer must first be installed in Windows before it can be used in the POS.

Step 1 – Connect the Printer:

43. Connect the printer to the Windows PC using a USB cable
44. Power on the printer

Step 2 – Install the Printer Driver:

45. Download the driver for your printer model from the manufacturer's website (e.g., Epson, Star Micronics, or Xprinter)
46. Run the driver installer and follow the on-screen instructions
47. When prompted, select USB as the connection type

Step 3 – Add the Printer in Windows:

48. Open Windows Settings → Bluetooth & devices → Printers & scanners

49. Click Add device and wait for Windows to detect the printer
50. Select your printer from the list and click Add device to complete the installation
51. Note the exact printer name as it appears in Windows — you will need this in the next step

Step 4 – Configure in Odoo:

52. Navigate to Point of Sale → Configuration → Point of Sales
53. Select your POS shop
54. Find the ESC/POS Printer field and enter the Windows printer name exactly as it appears in Printers & scanners
55. Click Save

The printer will automatically print the fiscal invoice after each validated transaction.

NOTE: If the printer does not print, open Windows Settings → Printers & scanners, select the printer, and click Print a test page to confirm the printer is working correctly in Windows before troubleshooting in Odoo.

6.0 Troubleshooting

6.1 Common Issues

Issue	Likely Cause	Solution
TaxCore signs invoice but SDC portal shows no record	referentDocumentNumber or invoiceType mismatch	Ensure original order was a Normal Sale (NS). Refunds of proforma orders may not appear on portal.
Barcode scanning does not find the product	Product has no Barcode or GTIN set	Open product → Product Information tab → fill in GTIN field and save
Payment method error on session open	Cash method already linked to another POS	Each register needs its own POS Cash method. Use POS Cash 2 for second register.
Certificate error on TaxCore call	PFX not converted or wrong password	Go to FRCS → V-SDC Settings → click Convert PFX button

6.2 Upgrade Commands

Use the following commands to upgrade specific modules after making changes:

Upgrade custom_pos:

```
"C:\Program Files\Odoo 18.0.XXXXXXXX\python\python.exe" "C:\Program Files\Odoo
18.0.XXXXXXXX\server\odoo-bin" -d FRCS -u custom_pos --stop-after-init
```

Upgrade pos_minicoa (payment methods):

```
"C:\Program Files\Odoo 18.0.XXXXXXXX\python\python.exe" "C:\Program Files\Odoo
18.0.XXXXXXXX\server\odoo-bin" -d FRCS -u pos_minicoa --stop-after-init
```

Upgrade l10n_fj_minicoa (currency/taxes):

```
"C:\Program Files\Odoo 18.0.XXXXXXXX\python\python.exe" "C:\Program Files\Odoo
18.0.XXXXXXXX\server\odoo-bin" -d FRCS -u l10n_fj_minicoa --stop-after-init
```

NOTE: Replace XXXXXXXX with your Odoo version number (e.g., 20260331) and FRCS with your database name. After any upgrade, restart the Odoo Windows service and press Ctrl+Shift+R in the browser.

7.0 Support and Contact

Contact	Details
Development Team	FRCS – IT Team
FRCS TaxCore Support	n.milojevic@frcs.org.fj
FRCS V-SDC Sandbox	https://vsdc.sandbox.vms.frcs.org.fj
FRCS V-SDC Portal	https://frontendui.test.taxcore.dti.rs
Odoo Documentation	https://www.odoo.com/documentation/18.0

For TaxCore certification issues, contact FRCS with your test invoice samples including at least two tax rates, two payment types, GTIN on items, and your POS number in the format accreditation_number/system_version (e.g., 182/1.0.0).

End of Configuration Manual

FreePOS | Fiji Revenue and Customs Service (FRCS) | FRCS